Table of Contents

..... 1
Problem 1 1
Problem 1.A: Time-Domain Signals 1
Problem 1.B: Frequency Response 2
Problem 1.C: Parameter Sweep 3
Problem 2.B 4
Problem 2.C 7
Problem 3: Phasor Analysis 8

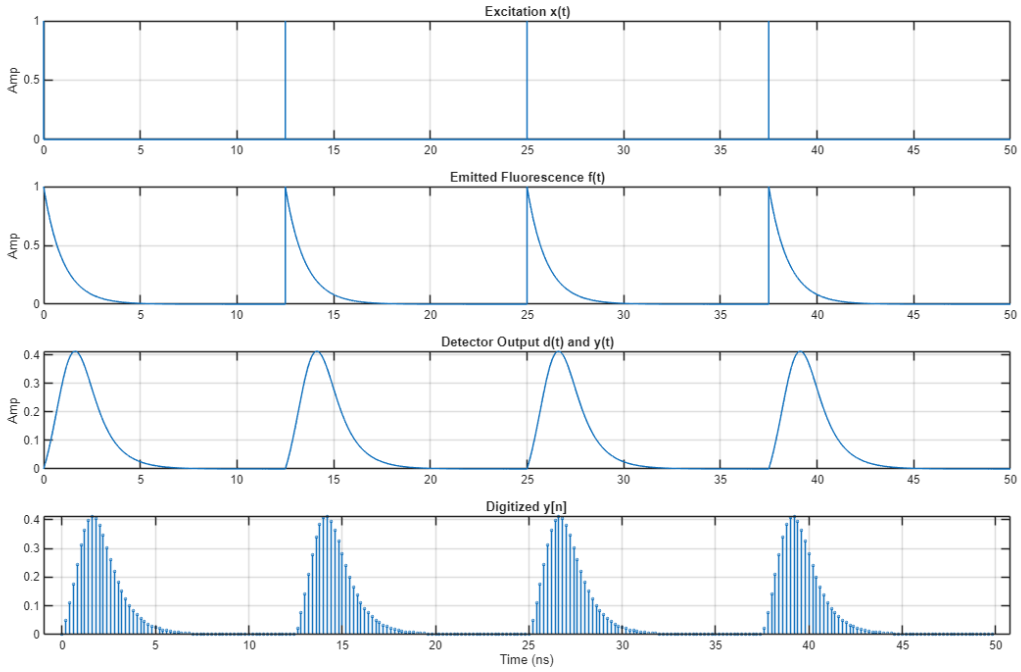

```
% Case Study 2

% Please run the code while you are within the source directory, otherwise
% the paths will not work.
```

Problem 1

```
tau = 1.0;       % ns
TL = 12.5;      % ns
Ts = 0.2;       % ns
t_end = 50;    % ns
sample_period = Ts/1000;

t = 0 : sample_period : (t_end - sample_period);
```

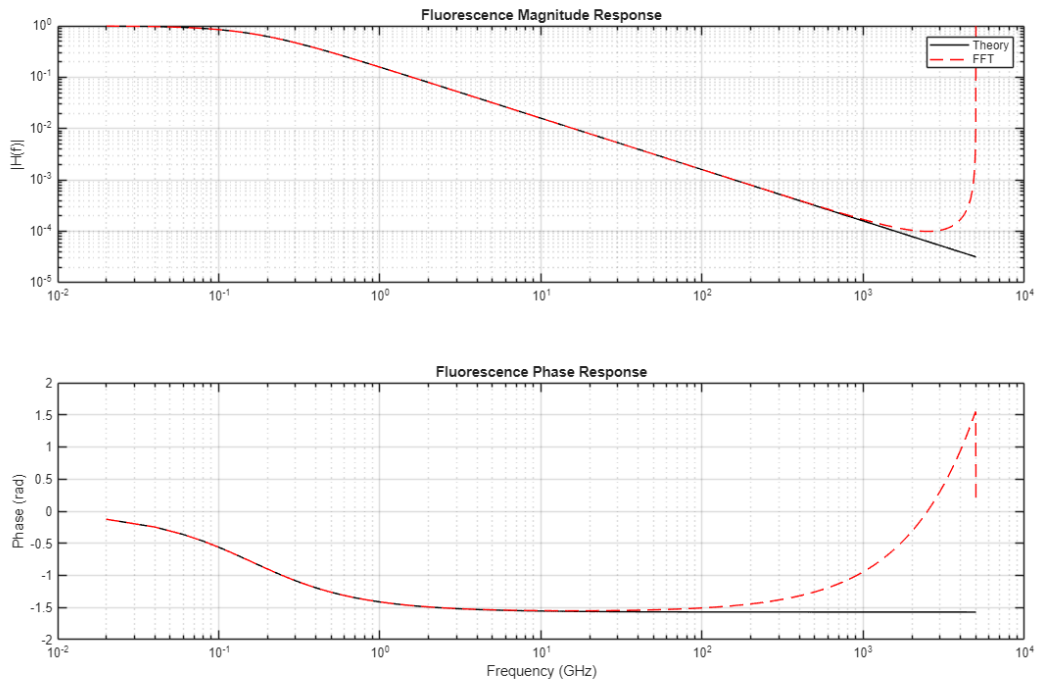
Problem 1.A: Time-Domain Signals

```
x = get_excitation(t, TL);
hf = get_fluorescence_irf(t, tau);

t0 = 1;
sigma = 1;
hd = get_detector_irf(t, t0, sigma);

f = ifft(fft(x) .* fft(hf));
d = ifft(fft(f) .* fft(hd));
[y_n, t_n] = get_sampling(t,d,Ts);

figure(1);
subplot(4,1,1); plot(t, x); title('Excitation x(t)'); ylabel('Amp'); grid on;
subplot(4,1,2); plot(t, f); title('Emitted Fluorescence f(t)');
ylabel('Amp'); grid on;
subplot(4,1,3); plot(t, d); title('Detector Output d(t) and y(t)');
ylabel('Amp'); grid on;
subplot(4,1,4); stem(t_n, y_n, 'MarkerSize', 2); title('Digitized y[n]');
xlabel('Time (ns)'); grid on;
```



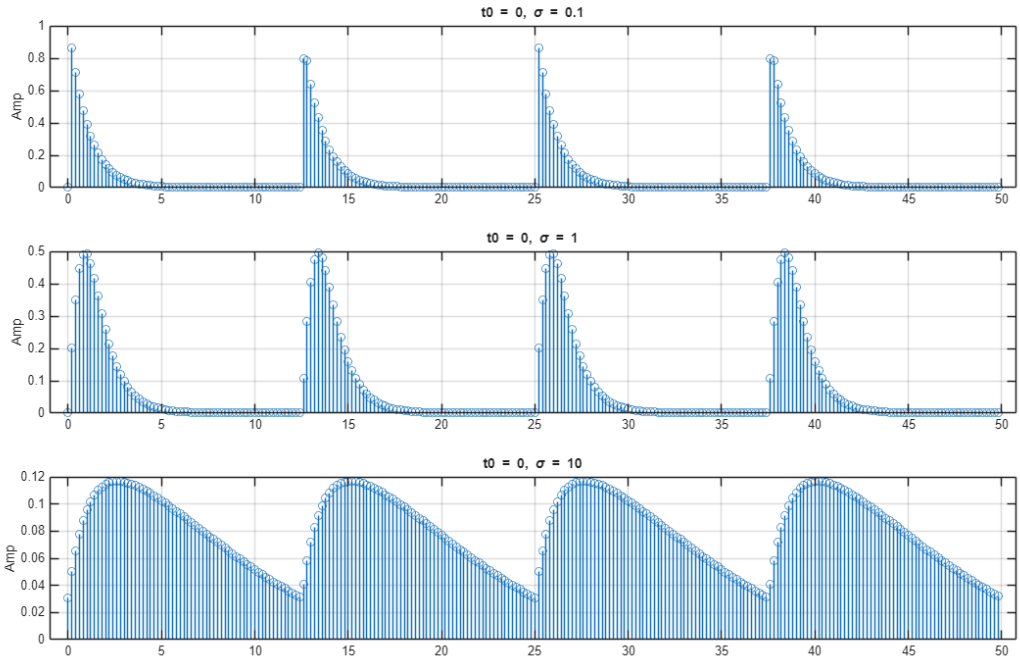
Problem 1.B: Frequency Response

```
fs = 1/sample_period;
N = length(hf);
freqs = (0:N-1)*(fs/N);

H_theory = tau ./ (1 + 1j * 2*pi*freqs * tau);

H_num = fft(hf) * sample_period;

figure(2);
subplot(2,1,1);
loglog(freqs, abs(H_theory), 'k', freqs, abs(H_num), 'r--');
title('Fluorescence Magnitude Response'); ylabel('|H(f)|');
legend('Theory','FFT'); grid on;
subplot(2,1,2);
semilogx(freqs, angle(H_theory), 'k', freqs, angle(H_num), 'r--');
title('Fluorescence Phase Response'); ylabel('Phase (rad)');
xlabel('Frequency (GHz)'); grid on;
```



Problem 1.C: Parameter Sweep

```
% Sweep over sigma and t0 values
sigma_values = [0.1, 1.0, 10];
t0_values = [0, 0, 0];
results = zeros(length(sigma_values), length(t0_values));

figure(3);

for i = 1:length(sigma_values)
    hd = get_detector_irf(t, t0_values(i), sigma_values(i));
    f = ifft(fft(x) .* fft(hf));
    d = ifft(fft(f) .* fft(hd));

    subplot(length(sigma_values), 1, i);
    [y_n, t_n] = get_sampling(t,d,Ts);
    stem(t_n, y_n);
    title(['t0 = ', num2str(t0_values(i)), ', \sigma = ',
num2str(sigma_values(i))]);
    ylabel('Amp'); grid on;
end
```

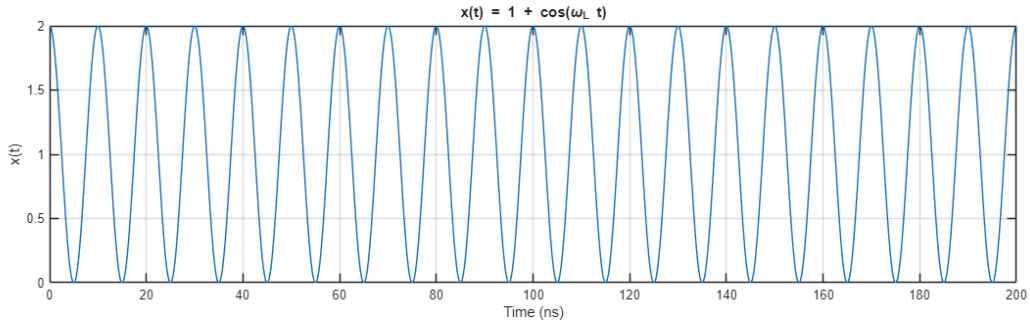


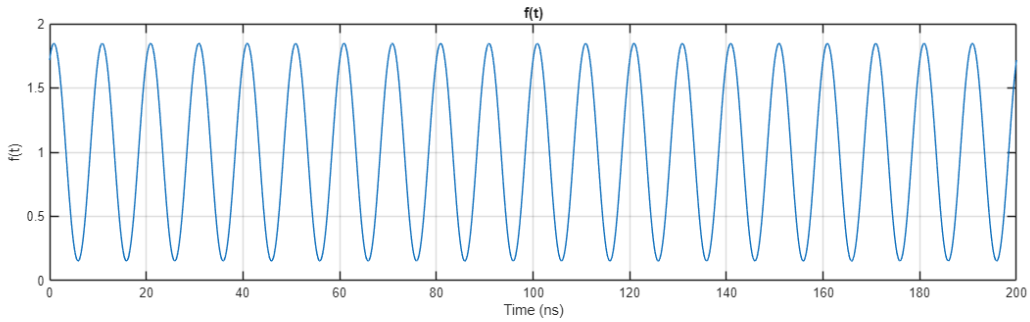
Problem 2.B

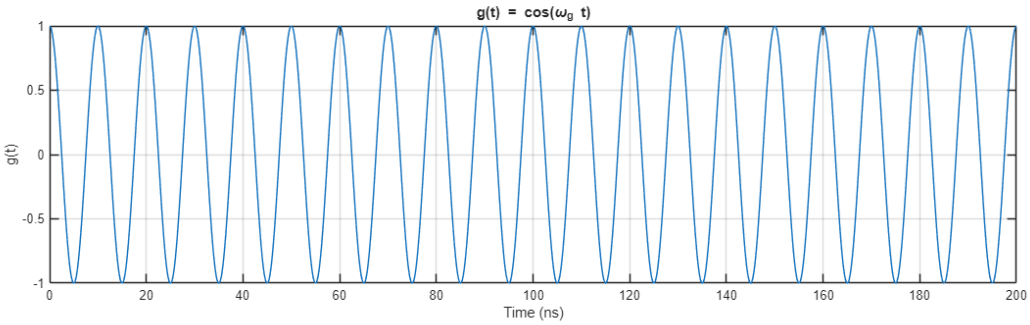
```
tau2 = tau;

f_L = 0.1;
w_L = 2*pi*f_L;

fs_ADC = 1e-4;
f_Nyq = fs_ADC/2;

% Choose g(t) to shift 100 MHz down to 10 kHz
f_LF = 1e-5;
f_g = f_L - f_LF;
w_g = 2*pi*f_g;
w_LF = 2*pi*f_LF;

M = 1 / sqrt(1 + (w_L*tau2)^2);
phi = atan(w_L*tau2);

t_rf = 0:0.02:200;

x_t = 1 + cos(w_L*t_rf);
f_t = 1 + M*cos(w_L*t_rf - phi);
g_t = cos(w_g*t_rf);
y_t = f_t .* g_t;

figure(4);
subplot(2,1,1);
```

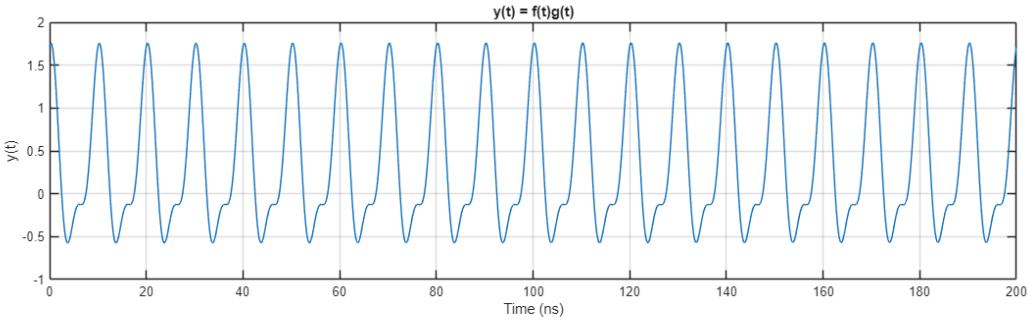
```
plot(t_rf, x_t);
title('x(t) = 1 + cos(\omega_L t)');
xlabel('Time (ns)');
ylabel('x(t)');
grid on;

subplot(2,1,2);
plot(t_rf, f_t);
title('f(t)');
xlabel('Time (ns)');
ylabel('f(t)');
grid on;

figure(5);
subplot(2,1,1);
plot(t_rf, g_t);
title('g(t) = cos(\omega_g t)');
xlabel('Time (ns)');
ylabel('g(t)');
grid on;

subplot(2,1,2);
plot(t_rf, y_t);
title('y(t) = f(t)g(t)');
xlabel('Time (ns)');
ylabel('y(t)');
grid on;

dt_fft = 1;
t_fft = 0:dt_fft:(1e7 - dt_fft);

f_fft = 1 + M*cos(w_L*t_fft - phi);
g_fft = cos(w_g*t_fft);
y_fft = f_fft .* g_fft;

N = length(t_fft);
fs_fft = 1/dt_fft;

Y = fft(y_fft)/N;

Nhalf = floor(N/2);
freq = (0:Nhalf)*(fs_fft/N);
Y_adjusted = Y(1:Nhalf+1);

Ymag = 2*abs(Y_adjusted);
Ymag(1) = abs(Y_adjusted(1));

freq_kHz = freq * 1e6;

figure(6);
plot(freq_kHz, Ymag);
xlim([0 100]);
title('Magnitude of Y(j\omega)');
xlabel('Frequency (kHz)');
```

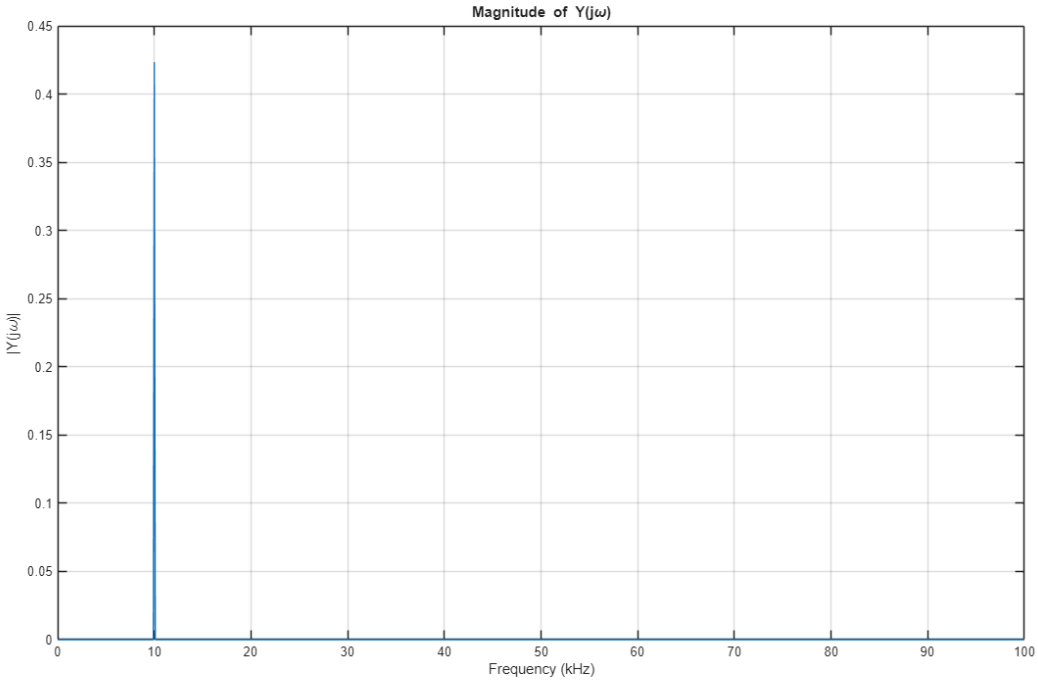
```
ylabel('|Y(j\omega)|');
grid on;
```











Problem 2.C

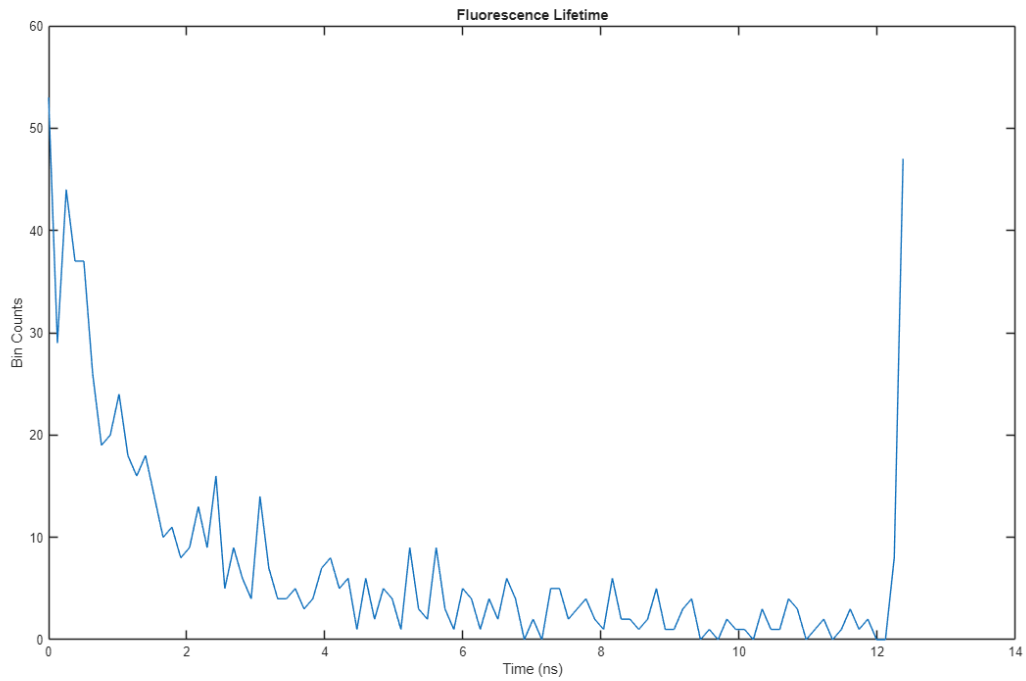
```
x_fft = 1 + cos(w_L*t_fft);
f_fft = 1 + M*cos(w_L*t_fft - phi);
g_fft = cos(w_g*t_fft);

c_fft = x_fft .* g_fft;
y_fft = f_fft .* g_fft;

C = fft(c_fft)/N;
Y = fft(y_fft)/N;

C_adjusted = C(1:Nhalf+1);
Y_adjusted = Y(1:Nhalf+1);

Cmag = 2*abs(C_adjusted);
Ymag = 2*abs(Y_adjusted);

Cmag(1) = abs(C_adjusted(1));
Ymag(1) = abs(Y_adjusted(1));

Cphase = angle(C_adjusted);
Yphase = angle(Y_adjusted);

[~, idx_LF] = min(abs(freq - f_LF));

A_copy = Cmag(idx_LF);
phase_copy = Cphase(idx_LF);
```

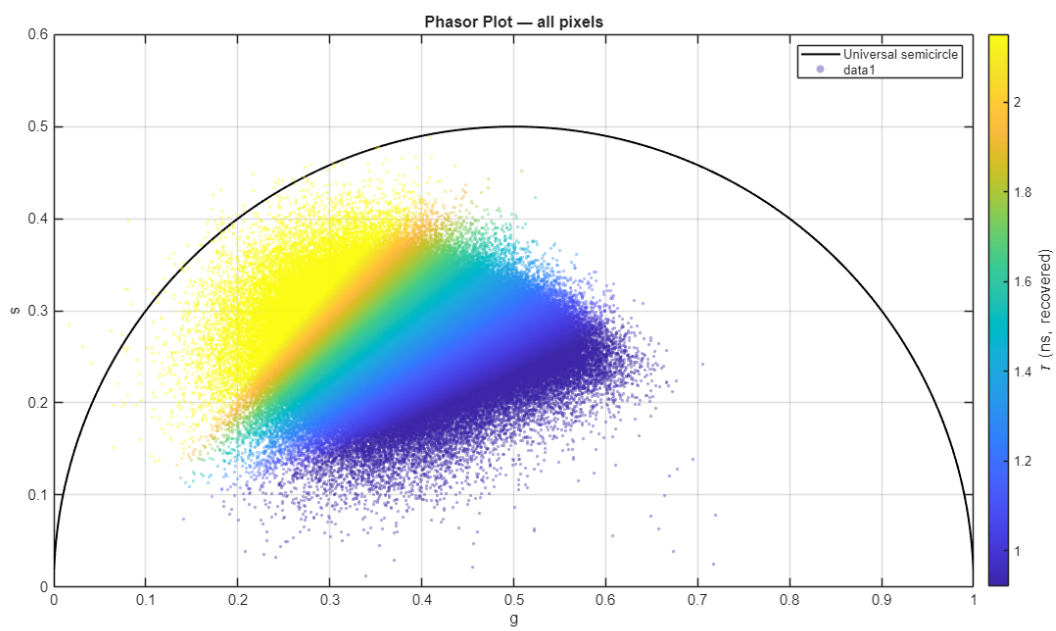
```
A_f = Ymag(idx_LF);
phase_f = Yphase(idx_LF);

M = A_f / A_copy;

delta_phase = angle(exp(1j*(phase_f - phase_copy)));

phi = -delta_phase;

tau_from_phase = tan(phi) / w_L;

fprintf('M = %.4f\n', M);

fprintf('Phase difference = %.4f rad\n', delta_phase);

fprintf('Lifetime from phase = %.4f ns\n', tau_from_phase);
fprintf('True lifetime = %.4f ns\n', tau2);
```

```
M = 0.8467
Phase difference = -0.5610 rad
Lifetime from phase = 1.0000 ns
True lifetime = 1.0000 ns
```

Problem 3: Phasor Analysis

```
i_info = load("../input/FLIMhistogram.mat");

t_limit = (0:97) * (TL/98);   % 1x98, matches 3rd dim of H
H = i_info.FLIMhistogram;

% mean decay across all pixels, find peak and circshift to align
mean_decay = squeeze(mean(mean(double(H), 1), 2));  % 98x1
[~, peak_bin] = max(mean_decay);
H_aligned = circshift(H, -peak_bin, 3);  % roll peak to bin 1

plot_phasor(TL, t_limit, H_aligned);
```





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